Lav Kumar Singh, Ph.D.

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www.researchgate.net/profile/Lav-Kumar-Singh
https://arxiv.org/a/kumar_l_2.html



Employment History

20.07.2026 – Present	**Assistant Professor,** Amrita Vishwa Vidyapeetham Deemed to be University, Haridwar Campus.
01.04.2025-31.03.2026	**Assistant with a doctoral degree (Post-doctoral),** Institute of Mathematics, Physics and Mechanics (IMFM), Ljubljana (Slovenia).
01.07.2024 – 31.03.2025	**Assistant Professor,** Madhav Institute of Technology and Science- Gwalior (M.P) India.
13.03.2023 – 30.06.2024	**Post-Doc Fellow,** Indian Institute of Science Education and Research, Bhopal (M.P).
23.12.2022 – 28.02.2023	**Research Associate,** Indian Institute of Technology, Hyderabad (Telangana).
25.07.2022 – 24.10.2022	**Research Associate,** Indian Institute of Technology, Gandhinagar (Gujrat).

Education

2017 – 2022	**Ph.D, Jawaharlal Nehru University** Thesis title: *On Arens Regularity of the projective tensor product of Schatten spaces.*
2014 – 2016	**M.Sc. Mathematics, Dr. Harisingh Gour Central University, Sagar (M.P)** . Grades (Percentage): 60%
2009 – 2012	**B.Sc. Honors Mathematics, Banaras Hindu University, Varanasi** . CGPA: 6.10

Research Publications

- L.K Singh, A. Peperko **Sherman-Takeda type theorems for locally C^*-algebras**, Arxiv.
- L.K Singh, A. Peperko, **Some Frechet spaces associated to Young functions satisfying Mulholland's inequality.** To appear in Filomat.
- L.K Singh, **Geometry of a Banach algebra $\mathcal{A}$ and the bidual of $L^1(G, \mathcal{A})$**, Journal of mathematical analysis and applications, Volume 533, Issue 1, 128056. 2024.
- L.K Singh, **Annihilators in the bidual of generalized group algebras**, Complex analysis and operator theory. Vol. 18, 68, 2024
- V.P Gupta, L.K Singh, **On strong Arens irregularity of the projective tensor product of Hilbert-Schmidt spaces**, Houston J. Math., Vol 48-2, 337-351. 2022.
- L.K Singh, **On Arens regularity of projective tensor producs of Schatten spaces**, Rocky Mount. J. Math. Vol. 51, issue 4, 1433-1441. 2021.
- L.K Singh, A.K Verma, **Some functorial properties of Schatten classes**. Adv. Oper. Theory 5, 1406–1413 (2020).

Conferences, Invited Talks and Research Visits

- Research visit at Czech Academy of Sciences, Institute of Mathematics (With Prof. Vladimir Müller), 20-28 March, 2026.
- Talk at "Seminar in algebra and functional analysis (SAFA)" at Faculty of Mathematics and Physics, University of Ljubljana, Slovenia. Talk title: Kaplanski Density property in locally C^*-algebras and a Sherman-Takeda type theorem.
- Contributory talk at SWOT-2025, University of Agriculture, Krakow (Poland). 21-24 July, 2025.
- IWOTA-2025, University of Twente, Enschede (Netherlands) 14-18 July, 2025.
- Contributory talk- International workshop on Operator theory and its Applications (IWOTA-2024)- University of Kent, Canterbury (U.K) (12-16 August, 2024).
- National conference on Operator theory and function spaces- Shiv Nadar Institute of Eminence (Attended 27-29 March, 2024).
- Indian Women in Mathematics conference- IISER Bhopal. (Attended 13 July-15 July, 2023)
- Invited as a short-term visiting scholar at department of mathematics IIT-Kanpur. (21 Nov-05 Dec)
- International conference on recent trends in mathematics, Hansraj college . Presented a paper. (22-24 Dec 2021, Online)
- Conference on algebra and analysis, Ambedkar University. Presented a paper. (23, April 2021, Online).
- Advanced instructional school in operator theory, IIT-Kanpur . (12 dec-28 dec, 2019)
- National workshop in functional analysis and topology, LNMIIT Jaipur. (March 2018)
- Annual foundation school-III, Kerala school of mathematics, Kozhikode.(July 2018)
- Annual foundation school-I, IIT-delhi. (December 2017)
- Summer research program, IISER-Mohali. Worked under guidance of Prof. I.B.S Passi and learned tropical algebras. (May 2013)

Teaching

2024-2025 Instructor/Lecturer at MITS- Gwalior for B.Tech course

- Research Methodology (PhD Coursework)
- M3: Engineering Mathematics-3 (B.Tech semester-IV)
- Linear Algebra (B.Tech Semester I)
- M2: Engineering Mathematics-2 (B.Tech semester-III)

2023-2024 Teaching Assistant for BS courses at IISER-Bhopal.

- MTH 101: Linear Algebra (Summer)
- MTH 201: Probability and Statistics (Winter)

Teaching (continued)

2019-2020 Teaching Assistant for M.Sc courses at Jawaharlal Nehru University.

- Complex Analysis (Summer)
- Functional analysis (Winter)

2018-2019 Teaching Assistant for B.Tech courses at Jawaharlal Nehru University, New Delhi.

- Engineering Mathematics- 1 (Summer)
- Engineering Mathematics- 2 (Winter)

Skills

Languages Strong reading, writing and speaking competencies for English and Hindi.

Coding Matlab, LaTeX.

Misc. Academic research, teaching, training, consultation, LaTeX typesetting and publishing.

Miscellaneous Experience

Awards and Achievements

2024 **SERB International Travel Support**, granted by Science and engineerign research board (Govt. of India) for presenting paper in IWOTA-2024, University of Kent, Cantebrury (U.K)

2017 **CSIR-JRF**, Fellowship granted by CSIR for pursuing PhD in Mathematics.

Certification

2016 **CSIR-NET for lecturership in Mathematical Sciences**. Awarded by Council for Scientific and Industrial research (Government of India).